Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Work out the value of $\dfrac{3^7 \times 3^{-2}}{3^3}$

$3^7 \times 3^{-2} \div 3^3 = 3^{7-2-3} = 3^2 = 9$

1 mark
Add powers when multiplying; subtract when dividing

1 mark
Evaluate the power 9

(Total for Question 1 is 2 marks)

2 $v^2 = u^2 + 2as$

$u = 12 \quad a = -3 \quad s = 18$

(a) Work out a value of v.

.....

(2)

(b) Make s the subject of $v^2 = u^2 + 2as$

.....

(2)

(Total for Question 2 is 4 marks)

DO NOT WRITE IN THIS AREA

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